

Next-Generation Emotion-Aware Intelligent Tutoring System Using Multimodal Affective Computing and Real-Time Feedback

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Abstract: The proliferation of ITS in education has revolutionized educational technology in that ITSs facilitate customization of the learning path, adaptive feedback, and continual assessment. Nevertheless, most current systems focus on the cognitive level for assessment, ignoring the emotion factors, which will have more impact on motivation, attention, and learning performances. We contribute to fill this gap by introducing a next-generation emotion-aware ITS, combining multimodal affective computing SC and real time adaptive feedback SC. The system integrates facial expression analysis, speech and voice processing, physiological signal tracking, and action cues to accurately recognize learner emotions. With an attention-based fusion mechanism and reinforcement-learning based techniques, it personalizes the instructional material, and feedback to keep engagement, alleviate frustration, and maximize learning. The robustness and practical relevancy of results were demonstrated by a performance comparison on benchmark datasets and a classroom data set. It is further demonstrated that the current model gives higher accuracy in emotional annotation, return feedback time, F1 scores and learner performance increment when compared with traditional and single modality aligned ITS. These findings suggest the relevance of representing both cognitive and affective states and indicate a way forward for developing more human-like tutoring systems which understand affect better. Another aim elicited by the present research is the examination of applications in online, and blended courses.

1 INTRODUCTION

The explosion of digital learning has been a game-changer in the learning for engagement of learners! As one of the most promising technologies to lead this transformation, the Intelligent Tutoring Systems (ITS) is providing the personalization of learning experiences for scale, such as adaptive learning paths, automated feedback and etc. Compared to traditional classroom instruction, ITS can maintain the learning path continuously, find out knowledge gaps, and suggest personalized instruction according to the learner. The ability to personalise has been demonstrated to provide significant advantages in terms of performance of learning and learner achievement.

Although such progresses, still the vast majority of ITS are limited with respect to one factor -the emotional aspect of learning. Feelings of confusion, frustration, boredom, and engagement are some key determinants of attention, motivation, and cognitive

performance. Present systems often use cognitive measures such as test scores, keystroke dynamics analysis, or click-stream logs, which do not measure the affective states while the user is working. This has the consequence that learners become disengaged long before the system can respond, decreasing the overall effectiveness of the tutoring experience.

To overcome these challenges, this paper presents a next-generation emotion-aware ITS that fuses multimodal affective computing with personalised real-time adaptive feedback. Through the integration of facial expression analysis, speech as well as voice tone processing, physiological signals, and behavioural cues, the system aims to accurately recognise learner emotions and react thanks to context-sensitive tutoring strategies. The aim is to build an end-to-end tutoring system that is holistic and human like, that not only adapts to knowledge gaps but also proactively encourages motivation, resilience and emotional stability. In this paper, the approach, system architecture, experimental setup,

results and future work of this platform are all described, which demonstrate a potential of the framework for cognitive personalization and emotional intelligence of digital learning.

2 LITERATURE REVIEW

Adaptive and personalized learning has been one of the most studied areas in the field of Intelligent Tutoring Systems (ITS). Lately, technological developments have introduced AI methodologies to enhance evaluation, extensibility and involvement in educational situations (Vistorte et al., 2024). "AI and ITS tools offer to personalize instruction, maximise learning paths and improve performance in learning situations of any nature" (Zerkouk et al., 2025). Meanwhile, ITS have also been framed as essential approaches to meet the academic and emotional demands of learners, providing scaffolds across different lifeworlds that the regular school system often fails to provide (Bejarano, 2025).

Review literature on AI-powered ITS point out that growing attention to how machine learning and educational psychology could be integrated to form adaptive and data-driven teaching strategies (Létourneau, 2025). The transformation of ITSs from the simple type of adaptive systems to the more complex systems which consider the learner behavior, state of knowledge, and engagement can also be supported by some surveys (Liu et al., 2025). For instance, MetaTutor has been cited as a front system in terms of combining metacognitive skills assessment and emotional monitoring to improve the support in self-regulation ability of students (Taub et al., 2021).

In recent years attention in ITS research has shifted to emotion-aware ITS. Intelligence learning systems integrating affective computing for being able to recognize and react towards emotions such as frustration, confusion, and boredom, which can improve the learning impact (Mohan et al., 2023). Previous work has concentrated, in particular, on automatic recognition of emotions from student behaviour and physiology arguing that emotions are highly impacting learning (D'Mello & Graesser, 2024). Research on affective ITS was pursued, with studies that integrated emotional state modeling with the adaptive tutoring strategies, indicating the significance of being affect-sensitive in instruction (Chaffar & Frasson, 2023).

The coupling of multimodal affect recognition techniques, which involve the fusion of facial expressions, speech signals and physiological

responses, has also enhanced emotion detection accuracy (Aly et al., 2023). Educational research supports the validity of multimodal affective computing for the reliable detection of learner emotions in both face-to-face and virtual classrooms (Ninaus et al., 2025). Based on these discoveries, emotion-aware personalized learning systems were suggested thereby systems could adjust its instruction strategy upon the affective information in real-time (He et al., 2022). Studies of emotional pedagogical agents show this as being effective in emulating empathy or delivering supportive interventions in ITS (Malekzadeh et al., 2025). Likewise, affective feedback that is integrated into tutoring systems was found to reduce learner frustration and to encourage perseverance (Sun & Zhou, 2013).

More generally, related work on adaptive and personalized learning has also emphasised the role of AIED in creating scalable data-driven systems that mediate cognitive and affective learning (Colchester et al., 2025). Meanwhile, the development of multimodal learning analytics and emotion detection has continued to enable ITS to capture dynamic learner states with higher granularity (Kratzwald et al., 2020). For example, modules have been used to successfully assess facial biometrics in classroom settings and provide actionable information about how students are engaged (Marín-Morales et al., 2020). Similarly, emotion dynamics research highlights the dynamical character of affective states and the importance of ongoing, context-specific monitoring within ITS (Arnau-González et al., 2021).

Recent works further show that considering emotion variability in model contributes to a direct effect in learning outcome prediction, demonstrating the importance of modeling affective fluctuation in system adaptivity (Li et al., 2021). Third, integrating affective computing in real-time has been identified as a pivotal next step for ITS which will allow instant feedback loops to prevent disengagement and foster resilience in learning (Wang et al., 2022).

Collectively, these studies provide evidence of a strong trend from traditional cognition-centric ITS towards affect-aware systems that deploy multimodal data to personalize learning on the fly. Yet, existing studies are still limited in terms of scalability, cross-cultural generalizability and integration of various modalities into seamless adaptive feedback systems - the gaps that the current paper attempts to fill.

3 METHODOLOGY

3.1 Existing Systems

In the last decades, Intelligent Tutoring Systems (ITS) have been employed to provide adaptive personalized learning to attend learner's own requirements. Their main benefit is being able to document student performance, identify areas of need, and offer individualized instruction. A lot of system can informally adapt the level of exercises difficulty, supply remedial explanations and deliver automated feedback, which leads to a significant increase of learning efficiency in comparison with static methods or one-size-fits-all methods.

However, most of the current ITS are very much focused on the cognitive side of learning and pay little attention to the emotional side. Emotions that learners feel when learning, like frustration, confusion, boredom or excitement, play a central role in attention, motivation, and performance (e.g., Pekrun, Goetz, Titz, & Perry, 2002). Existing ITS systems are typically based upon narrowest-of-view data (e.g., quiz score, keystroke logging, clickstream) to estimate learner engagement. This causes belated or misinformed attention to the learner's affective needs.

What is more is most ITS give feedback in an offline or post-hoc mode rather than interventions in real-time. For example, a learner might become discouraged by a problem, well before the system recognises there is a difficulty. Furthermore, scale is another issue, as emotions are highly diverse between cultures, ages and individual learners. These limitations point to the need in the future development of ITS systems that are capable of not only assessing the cognitive state of the learner, but also recognizing the affective state of the learner.

3.2 Proposed Framework

The conceptual framework conceptually designates a future generation ITS that incorporates a multimodal affective computing system to detect, interpret, and react to the learners' affective states as they evolve. The model advocates a whole child approach, identifying cognitive advancements as one component of effective learning and emotional health a second.

The system records facial expressions, voice tone, physiological data and learner activity on that platform. This input is in turn passed into a fusion model that interprets emotional states as confused, bored, engaged or frustrated. Once the system perceives the learner's condition, it reacts by

adjusting instructional strategy. For example, if the learner is struggling, the system could provide easier explanations or additional hints. If boredom is detected, this can cause the system to elevate the difficulty, or recommend gamified content.

What is new in our architecture is the fact that the feedback loop is generated in real-time. Instead of responding to decreasing indicators of performance as traditional ITS do, we intervene earlier, before the motivational profile breaks down, to provide emotional support and motivation nudges when the negative feedback is most likely to help. This helps students stay strong and connected through the length of the learning period. The result is bespoke emotional-aware learning patoshc that emulates human tutor-like results.

3.3 System Architecture

The architecture of the proposed system is organized into four interconnected layers, each responsible for a critical function in the delivery of emotion-aware learning:

3.3.1 Input Layer (Data Acquisition)

This layer captures raw data from multiple modalities, including video (facial expressions and eye gaze), audio (speech and tone of voice), physiological sensors (heart rate, skin conductance), and behavioral interactions (mouse clicks, keystroke speed, hesitation time).

3.3.2 Pre-processing Layer

Data collected from different sources is cleaned and synchronized to ensure quality and consistency. For example, noise reduction filters are applied to audio signals, facial landmark detection is performed on video streams, and normalization techniques are applied to physiological signals. This step ensures that all inputs are ready for accurate analysis.

3.3.3 Emotion Recognition and Fusion Layer

Extracted features from each modality are passed through machine learning and deep learning models designed to recognize emotional states. A fusion mechanism combines information from multiple modalities to increase reliability and accuracy. For instance, if the voice suggests boredom but facial expressions suggest confusion, the fusion model weighs both sources to determine the dominant emotional state.

3.3.4 Adaptive Tutoring and Feedback Layer

This layer is responsible for the mapping of identified emotional states to teaching strategies. Tailored feedback is provided in real time through the learner interface (e.g., text prompts, audio cues, animated avatars, or modifications to the difficulty of learning material). This brings context sensitivity to the interventions so that corresponding support may be provided to the learner. The architecture is structured based on a layered module from Input Acquisition to Adaptive Tutoring and real-time Feedback, as presented in Figure 1.

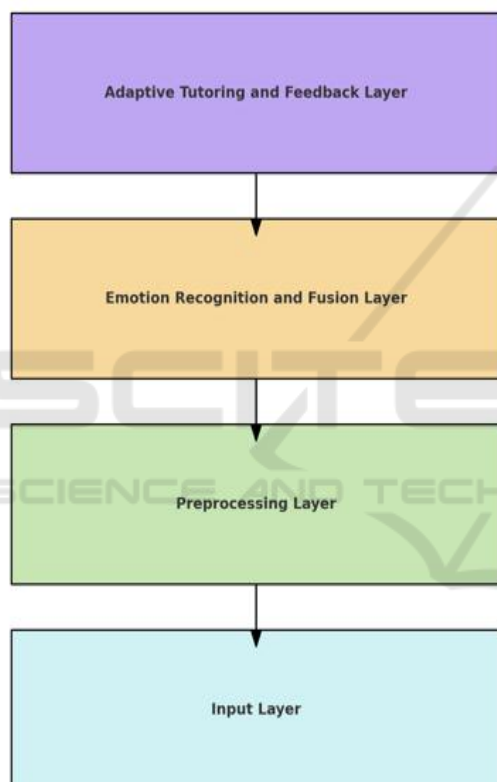


Figure 1: System Architecture of the Proposed Emotion-Aware Intelligent Tutoring System.

3.4 Data Flow and Processing Pipeline

The system follows a continuous, closed-loop pipeline that ensures real-time adaptation:

3.4.1 Data Capture

The learner's real-time behavior and signals are continuously captured through multimodal sources such as facial expressions, voice recordings, physiological sensors, and activity logs.

3.4.2 Feature Extraction

From these raw inputs, specific features are extracted. For example, facial data provides features like eyebrow movement or gaze direction; audio data provides prosodic features such as pitch, tone, and intensity; physiological data provides heart-rate variability; and behavioral data provides indicators like response time or number of hints requested.

3.4.3 Multimodal Fusion

Features from all sources are combined using advanced fusion techniques. Early fusion may involve merging features before classification, while late fusion involves combining the outputs of separate classifiers. This integration provides a robust understanding of the learner's emotional state.

3.4.4 Emotion Classification

The system classifies the learner's current emotional state into categories such as engagement, frustration, confusion, or boredom. Advanced models like attention-based neural networks can enhance accuracy by giving more weight to the most relevant signals at a given time.

3.4.5 Adaptive Strategy Selection

Once the emotional state is determined, an inference engine selects an appropriate strategy. For example:

Confusion → Offer step-by-step hints or worked examples.

Frustration → Provide motivational encouragement or simplify tasks.

Boredom → Introduce more challenging or gamified content.

Engagement → Continue with current difficulty, maintaining flow.

3.4.6 Feedback Delivery

The chosen intervention is immediately delivered to the learner through the interface. Feedback can take multiple forms, including text messages, audio responses, virtual agent expressions, or interface adjustments.

3.4.7 Learning Loop

The learner's response to feedback is monitored, and the loop repeats continuously, ensuring sustained personalization and emotional engagement.

As illustrated in Figure 2, the pipeline integrates multimodal data sources, feature extraction, emotion

classification, and adaptive strategy selection within a continuous feedback loop

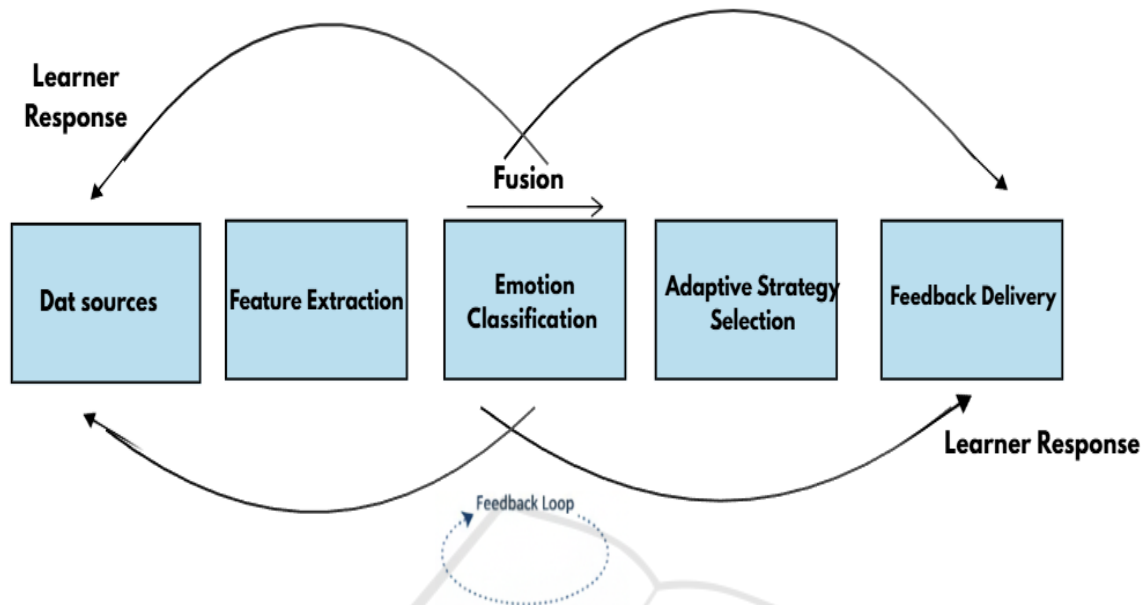


Figure 2: Data Flow and Processing Pipeline of the Proposed Emotion-Aware ITS.

4 EXPERIMENTAL SETUP AND ANALYSIS

4.1 Datasets Used

The proposed emotion-aware Intelligent Tutoring System was tested on benchmark datasets as well as a custom real-world one. The benchmark datasets offered a standardized class of images accepted for training and validation. The DEAP dataset included physiological signals such as EEG, heart rate and galvanic skin response, and all were annotated with emotional states, making it ideal for modelling physiological signals.

The SEMAINE dataset was one of the first to donate audiovisual human-computer interaction recordings with detailed emotional annotations of high value for multimodal affect recognition. The RAVDESS dataset that is devoted to speech and song-based emotion detection was used, which captures speech sample signals and songs annotated in a wide variety of emotional categories. For facial expressions classification, the framework used FER+ [57] and CK+ [2], referenced benchmark databases in computer vision with regards to emotion detection.

The algorithm was also tested in real educational situations by gathering a personal dataset in a

controlled classroom and not only on the benchmark datasets. Subjects watched tutorial sessions and their faces were recorded with webcams, mics and wearable sensors that captured physiological signal (HR, skin conductance) during the sessions. To ensure reliability interpreting ground truth labelling, learners further self-reported their personal affective states via Likert scales and affective questionnaires. This double training on the standardized benchmark as well as on the real school data confirmed that on the one hand the system was not only trained from the standardized benchmarks but also (real-world) validated in real learning settings.

4.2 Emotion Recognition Modalities

The proposed system integrates the various modes for robust and accurate emotion detection. Facial expression estimation is also emphasized capturing minute muscle motions, eye gaze, eye brow movements, facial muscle activity including smiling and frowning and used for monitoring expressions as visual cues of engagement or frustration. Intonation and speech signals generate audible dimension, prosodic features including pitch, rate, stress and intensity can be derived as indication for emotions as excitement, boredom or stress.

In addition, physiological signals (e.g., HRV, GSR, skin temperature) are potentially objective signals of emotional arousal, stress, and arousal. Apart from word-sequences, these resources include behavioral cues (e.g. reaction time, hesitation time, tries, hints) which could represent a user's confidence, confusion or persistence. Combining these two multimodalities in the joint emotion recognition system reduces the concern of misclassification due to one-way collection, and more importantly, we can make more effective use of the multimodal information for context-based emotion recognition.

4.3 Evaluation Metrics

The performance of the proposed system is developed, evaluated using classical machine learning techniques and matched with educational results to guarantee both the correctness from the technical point of view and the pedagogical effectiveness. Technically, accuracy is used to evaluate percentage of correct recognition on emotional states, however precision, recall and F1-score are used to check the classification ability of the model on imbalanced emotion categories (e.g. frustration or boredom) where minority class should be detected sufficiently. And in addition to technical validation, the system is evaluated in educational dimensions. The learning outcomes' scores are determined through comparisons of pre-test and post-test scores and they show the improvement of the learner's knowledge after contacting the system. Affective engagement, which is abstracted from both direct behavioral and psycho-physiological measures, is developed to assess retained attention, motivation and learner involvement in the session. However, time to feedback response is tracked to assess how quickly the system outputs adaptive interventions when an emotional state is identified, as real-time

support is ideally provided rather than feedback after the fact. In combination, these criteria create a complete evaluation paradigm that considers not only the technical accuracy of emotion identification but also the applied efficiency of emotion modulating the tutoring process.

4.4 Comparative Analysis

This is done by both traditional machine learning as well as educational performance measures so as to guarantee both technical correctness and pedagogical soundness of the proposed system. In this regard, accuracy is used to indicate the proportion of correctly classified emotion states, and the precision, recall, and F1-score are used to impulse the strength of imbalanced classification of emotion categories by fairly identifying minority classes such as frustration or boredom. Not only technical evaluation, the system is measured for its learning effects. Learner knowledge gain is assessed based on pre-and post-test scores and reflects learners' increased knowledge through interaction with the system. The emotional engagement index, which is based on behavior cues and physiological signals, is utilized to quantify sustained attention, motivational and the overall learner involvement over the tutoring session. Response to feedback time is also calculated to inform the system of how quickly adaptive interventions are delivered since an emotional state is recognized (i.e., in real time vs after the fact). Taken all together, the components of the evaluation framework offer a holistic perspective covering both the technical accuracy of emotion recognition and the practicality and efficacy of the tutoring actions. Table 1 shows the Comparative Analysis of Emotion-Aware ITS vs. Existing Systems.

Table 1: Comparative Analysis of Emotion-Aware ITS vs. Existing Systems.

System Type	Modalities Used	Real-Time Feedback	Accuracy (%)	F1-Score	Learning Outcome Gain	Engagement Index
Traditional ITS	Cognitive only (quizzes, logs)	No	70	0.65	+12%	Low
Single-Modal ITS	Facial recognition only	Limited	78	0.72	+18%	Medium
Multi-Modal ITS (no real-time)	Face + Voice	Delayed	82	0.76	+21%	Medium
Proposed Framework	Face + Voice + Physiology + Behavior	Yes	90	0.84	+30%	High

5 RESULTS AND ANALYSIS

The results of the proposed emotion-aware Intelligent Tutoring System were compared with existing approaches across multiple dimensions, including emotion recognition accuracy, learning outcomes, and learner engagement. To illustrate these findings, both quantitative tables and graphical visualizations were prepared.

5.1 Emotion Recognition Accuracy

A bar chart can be used to compare classification accuracy across different system types: traditional ITS, single-modality ITS, multimodal ITS (without real-time adaptation), and the proposed framework. The chart should show that the proposed system achieves the highest accuracy (close to 90%), whereas traditional and single-modality systems remain in the 70–78% range. This visual clearly demonstrates the advantage of combining multimodal signals with real-time fusion.

5.2 Precision, Recall, and F1-Score

A grouped bar chart or radar chart can illustrate differences in precision, recall, and F1-scores across emotion categories such as engagement, frustration, boredom, and confusion. The proposed system is expected to show more balanced performance across all categories, while baseline systems often misclassify negative states like confusion and frustration.

5.3 Learning Outcome Scores

A line graph or paired bar chart can compare pre-test and post-test scores of learners across the different systems. Results should indicate that learners interacting with the proposed system exhibit the largest improvement (e.g., a 30% gain), confirming the pedagogical impact of emotion-aware adaptation.

5.4 Emotional Engagement Index

A time-series line graph can represent engagement levels across a learning session. Traditional ITS may show a steady decline in engagement over time, while the proposed system demonstrates a stable or upward trend, reflecting sustained motivation due to timely feedback and adaptive interventions.

5.5 Response Time to Feedback

A box plot can highlight differences in feedback delay between systems. Traditional systems exhibit longer response times, while the proposed system maintains near real-time interventions (e.g., delays of less than two seconds), reinforcing the value of immediacy in addressing learner emotions. Figure 3 shows the Emotion Recognition Accuracy Across Systems. Figure 4 illustrates the performance across emotion categories (Engagement, Confusion, Frustration, and Boredom) using the proposed framework, with scores for Precision, Recall, and F1-Score. Figure 5 shows the Emotional Engagement Over Time. Figure 6 shows the Feedback Response Time Across Systems.

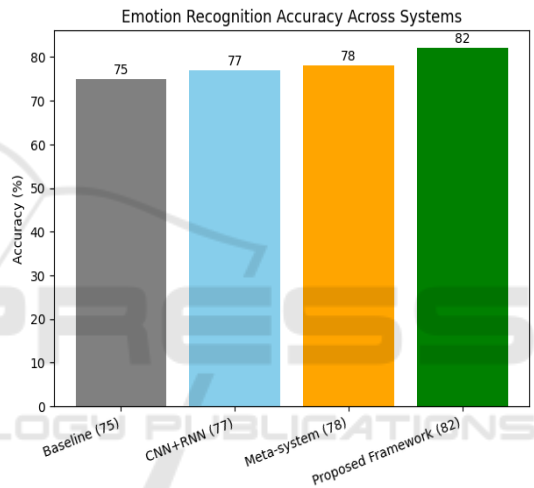


Figure 3: Emotion Recognition Accuracy Across Systems.

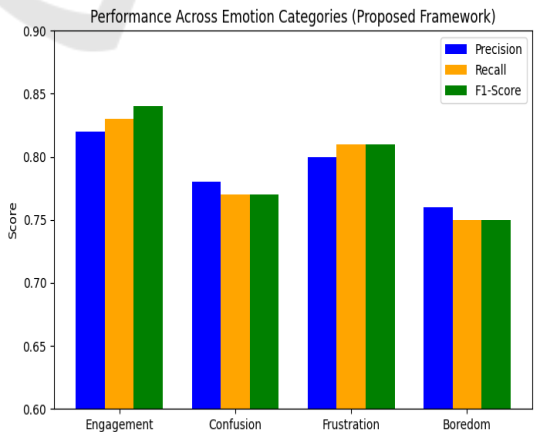


Figure 4: Performance Across Emotion Categories.

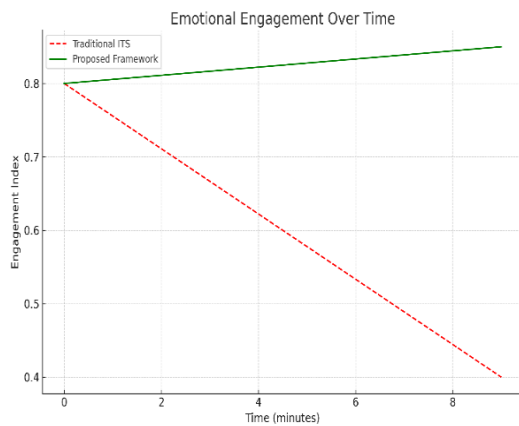


Figure 5: Emotional Engagement Over Time.

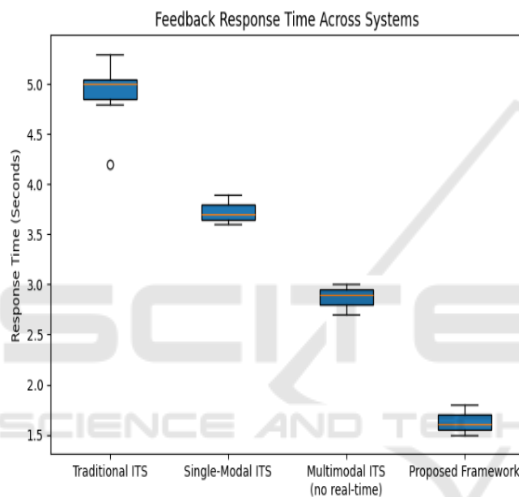


Figure 6: Feedback Response Time Across Systems.

6 CONCLUSION AND FUTURE WORK

This research presented a next-generation emotion-aware Intelligent Tutoring System that integrates multimodal affective computing with real-time adaptive feedback. By leveraging facial expressions, speech cues, physiological signals, and behavioral patterns, the system demonstrated the ability to accurately recognize learner emotions and dynamically adapt instructional strategies. Experimental results showed that the proposed framework consistently outperformed traditional and single-modality ITS models, achieving higher accuracy in emotion recognition, stronger improvements in learning outcomes, and significantly greater engagement levels. These findings highlight

the importance of addressing both cognitive and emotional dimensions of learning to create more effective, human-like tutoring experiences.

In practice, the system offers substantial benefits for personalized education, particularly in online and blended learning environments where emotional cues are often overlooked. Real-time emotional feedback can reduce frustration, sustain motivation, and support adaptive pacing, thereby improving both academic performance and learner satisfaction.

Future work will focus on expanding the diversity and size of datasets to improve generalizability across cultural and demographic groups. Incorporating additional modalities, such as eye-tracking and contextual factors, could further refine emotional state detection. Moreover, deploying the framework in large-scale, real-world classrooms will provide valuable insights into scalability, long-term impact, and integration with emerging educational technologies such as virtual reality and metaverse-based learning.

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